

Open the Model: Explain, Challenge, Debug

EXPLANATION IS EVIDENCE NOT CERTIFICATION

The selected model will be investigated, not blindly trusted.

OBJECTIVE

The investigation path

Previous: a legitimate model was selected. Now: its behavior must be inspected. Next: decisions will be controlled.

Final artifact

AI Explanation Casebook

Evidence target

Global reliance + local cases

Decision target

One wrong case -> one next test

Selected model

Global importance

Local cases

Wrong case

Hypothesis

Next test

The session is a model investigation, not a chart gallery.

A plausible explanation can still describe a wrong prediction

Previous: scores chose the model. Now: one prediction is put on trial.

Case

Actual: MEDIUM
Predicted: HIGH
Confidence: 0.72

First thought

High drought + heat might make HIGH risk plausible.

Scientific caution

Plausible domain reasoning is not yet a model explanation.

Question: what did the fitted model actually use?

From map to street view to accident report

The story rises from overall behavior to one wrong case.

GLOBAL

Which features affect performance across many cases?

LOCAL

What moved this one prediction toward its selected class?

DEBUGGING

What should be tested after one wrong prediction?

many cases

one case

one mistake

next experiment

The notation used for the whole session

This page prevents equations from appearing without context.

Symbol	Meaning	Plain language
x	one input record	one region-month case
x_j	feature j in x	e.g., drought_index for this case
c	class index	LOW, MEDIUM, or HIGH
$c^*(x)$	predicted class	the class with largest predicted probability
$f_c(x)$	model output for class c	score/logit or probability depending on explainer
S	evaluation score	Macro F1 in this session
$\phi_{j,c}$	SHAP value	feature j contribution toward class c

Why Macro F1 still matters before explainability

Previous sessions selected a legitimate model. These scores are the baseline for inspection.

$$F_{1,c} = \frac{2P_cR_c}{P_c + R_c} = \frac{2TP_c}{2TP_c + FP_c + FN_c}$$

Symbols: P_c =precision for class c ; R_c =recall; TP, FP, FN are class-specific counts.

Why it matters: F1 exposes whether each class has both usable precision and usable recall.

$$F_{1,\text{macro}} = \frac{1}{C} \sum_{c=1}^C F_{1,c}$$

Symbols: C =number of classes; $F_{1,c}$ =F1 for class c .

Why it matters: Macro F1 gives LOW, MEDIUM, and HIGH equal voting power.

Explainability does not replace validation. It starts after a valid model has already been selected.

Always explain the output that was actually chosen

Without this, a local explanation can answer the wrong question.

$$c^*(x) = \arg \max_{c \in \{1, \dots, C\}} \hat{p}_c(x)$$

Symbols: $\hat{p}_c(x)$ =predicted probability for class c ; argmax selects the largest one.

Why it matters: The explanation must target the predicted class $c^*(x)$, not a random class slice.

Interpretation rule

If a case is predicted MEDIUM, explain the MEDIUM output first. Explaining HIGH would describe a different score.

probabilities

predicted
class

class-specific
explanation

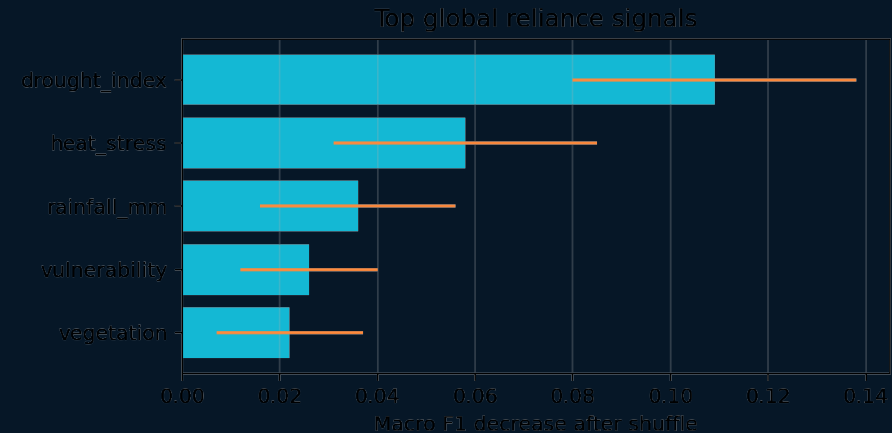
Permutation importance: shuffle, score, compare

Now the model is questioned at the dataset level.

$$I_j = \frac{1}{R} \sum_{r=1}^R [S_0 - S_{j,r}^{\text{perm}}]$$

Symbols: I_j =importance of feature j ; R =repeats; S_0 =baseline Macro F1; S^{perm} =score after shuffling.

Why it matters: If shuffling feature j repeatedly lowers Macro F1, the fitted model relied on that feature for this evaluation set.



Important boundary

Permutation importance is model-specific and sample-specific. It is not a causal claim and not the intrinsic value of the feature.

What should be said when the table appears

This turns a dataframe into a scientific sentence.

Feature	Mean decrease	Std.	How to say it
drought_index	0.1088	0.0291	Largest measured reliance
heat_stress	0.0580	0.0270	Secondary thermal signal
rainfall_mm	0.0364	0.0204	Useful but partly redundant
negative value	e.g., -0.005	small	Noise/redundancy; not physical harm

Say: “When this feature was disrupted, Macro F1 dropped by this amount.”

One case: replace one feature with a training reference

Now the investigation zooms from dataset behavior to one output.

$$\Delta_{j,c^*} = \hat{p}_{c^*}(x) - \hat{p}_{c^*}(x_{j \leftarrow b_j})$$

Symbols: Δ_{j,c^*} =local effect; $\hat{p}$ =predicted probability;
 b_j =training median reference.

Why it matters: A positive Delta means the original feature value supported the predicted class compared with the reference value.

LOCAL OCCLUSION PSEUDOCODE

```
base = predict_proba(sample)[class_id]
for feature in MODEL_FEATURES:
    altered = sample.copy()
    altered[feature] = training_median[feature]
    effect = base - predict_proba(altered)[class_id]
```

Output sentence

“Replacing drought_index with a typical training value lowered HIGH probability, so the original drought_index value supported HIGH.”

Probability change becomes a ranked local explanation

This is intentionally explained before SHAP so the intuition is clear.

Feature replaced	Original P(HIGH)	New P(HIGH)	Effect
drought_index	0.72	0.47	+0.25
heat_stress	0.72	0.55	+0.17
water_balance	0.72	0.60	+0.12
humidity	0.72	0.77	-0.05

Positive effect

The current value increased support for the selected class versus the reference.

Negative effect

The current value reduced support for the selected class versus the reference.

This is not a full game-theoretic attribution; it is a fast model-agnostic sensitivity test.

A prediction is treated like a team result

Now the attribution becomes more formal.

Baseline

Average model output before seeing this case.

Players

Each feature is a player that may join the information team.

Credit

SHAP distributes the movement from baseline to this output.

baseline output

feature contributions

case output

The formal credit allocation equation

This is where the symbols connect to the story.

$$\phi_{j,c} = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|! (M - |S| - 1)!}{M!} [v_c(S \cup \{j\}) - v_c(S)]$$

Symbols: F =set of all features; M =number of features; S =subset not containing j ; $v_c(S)$ =model output for class c when only S is known; $\phi_{j,c}$ =feature j credit.

Why it matters: Every subset S is a possible “before feature j joins” team; the equation averages feature j ’s marginal contribution across all possible teams.

Delivery sentence

“SHAP is not magic. It is a systematic averaging of how much a feature changes the model output when added to many possible feature coalitions.”

Baseline plus feature contributions reconstructs the model output

This is the equation students should remember.

$$f_c(x) = \mathbb{E}[f_c(X)] + \sum_{j=1}^M \phi_{j,c}$$

Symbols: $f_c(x)$ =model output for class c ; $\mathbb{E}[f_c(X)]$ =background baseline; $\phi_{j,c}$ =feature j contribution.

Why it matters: The explanation is complete in the model output space: baseline plus all SHAP values equals the explained output.

Critical warning

For classifiers this output may be a raw score or log-odds, not a direct probability. Do not say “percentage points” unless the explainer is explicitly in probability space.

Proper wording: “moved the class score upward.”

One case has one explanation vector per output class

The page connects SHAP tensors to LOW/MEDIUM/HIGH decisions.

Tensor axis	Meaning	Example
sample	which record	case 0
feature	which input dimension	water_balance
class	which model output	LOW / MEDIUM / HIGH

Rule

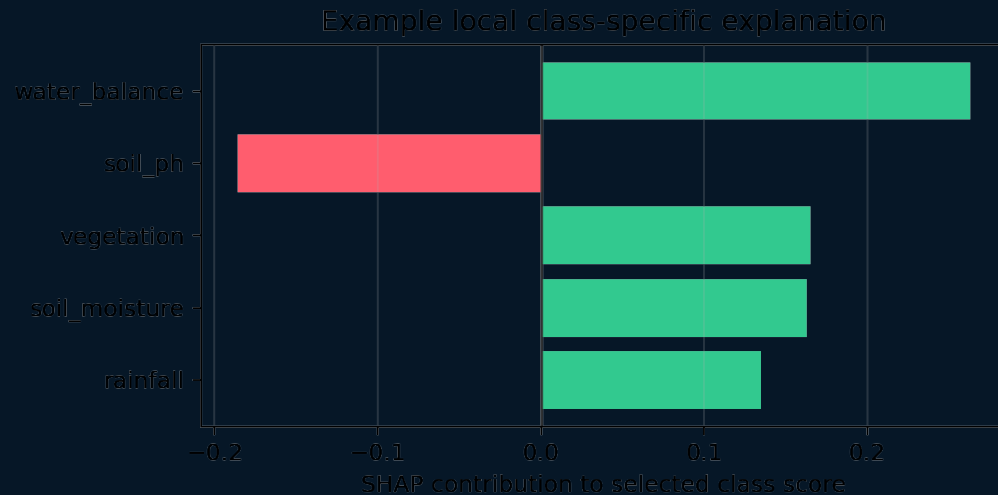
Extract values[case_index, :, predicted_class_id]. Otherwise the explanation may describe an output the model did not select.

Interpretation

Positive SHAP -> toward the selected class score. Negative SHAP -> away from that class score.

What to say when the local table appears

Now the class-specific explanation is translated into evidence.

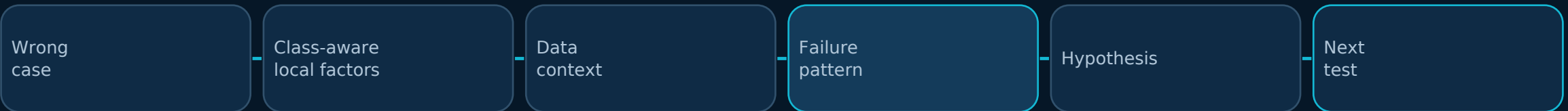


Column	How to explain it
feature	which signal is being credited
transformed_value	value after preprocessing
shap_value	direction and size of contribution
absolute_shap	ranking strength only

Do not say: “water_balance caused MEDIUM.” Say: “it increased the MEDIUM class score.”

Correct predictions reassure; errors teach

The story now uses explanations to choose a next experiment.



Question 1

Which features most increased support for the predicted class?

Question 2

Which expected signal was weak, absent, or contradictory?

Question 3

Does it suggest class overlap, missing signal, or feature limitation?

From explanation evidence to a testable action

An explanation becomes useful only when it changes what is investigated next.

Weak

“The model should be improved.”
Not testable.

Better

“The model may over-count
correlated drought features.”

Falsifiable

“Removing one redundant drought
feature should reduce
MEDIUM-to-HIGH errors without
reducing HIGH recall.”

Decision rule

A good next test specifies what changes, what stays frozen, which metric is evaluated, and what result would support or reject the hypothesis.

Would the same top factors appear under another background?

This prevents over-trusting one local explanation.

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Symbols: A=top factors from background A; B=top factors from background B; intersection=shared factors; union=all unique factors.

Why it matters: Jaccard overlap measures how similar two top-feature sets are. A low value warns that the explanation ranking is background-sensitive.

Example

A={drought, heat, water, humidity, vegetation}

B={drought, heat, water, humidity, disease}

J=4/6=0.667

Stable enough to discuss; not stable enough to overclaim.

Product model and tree candidate are not conflated

The selected product model is explained first; the tree model is a separate demonstration.

Product path

Selected logistic model
Global: permutation importance
Local: occlusion + Linear SHAP

Tree demonstration

Correct XGBoost candidate
Direct TreeExplainer
Limited to 30 cases for live speed

Why this matters

Explaining a different model does not explain the deployed decision mechanism. Specialized explainers are used when their model assumptions match.

How each output should be explained

The notebook now includes explanation prompts after major outputs.

Output	What it proves	What not to claim
Metric table	selected model performance	not causal behavior
Permutation table	global reliance on holdout	not intrinsic feature value
Local occlusion	probability sensitivity for one case	not complete attribution
SHAP table	class-specific score contributions	not real-world cause
Wrong case	debugging target	not a failure of the whole course
Jaccard	top-factor stability	not universal truth

Explain the model, then debug one mistake

The output is a casebook, not a chart dump.

1. Global

Permutation importance using Macro F1.

2. Local

One explained case per predicted class where available.

3. Error

One wrong prediction autopsy.

4. Stability

Two backgrounds + Jaccard.

5. Decision

Falsifiable hypothesis + next test.

Result

AI Explanation Casebook.